GROUP NO. 04

UNIVERSITY OF COLOMBO

DEPARTMENT OF MATHEMATICS

FM 3015 - CASE STUDY IN FINANCE

Analysis of Financial Strength and Stock Market Health of LOLC Holdings

Group Members

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1. Introduction

Founded in 1980, LOLC Holdings has grown into Sri Lanka's largest diversified conglomerate. Companies operating across multiple industries face constant pressure to maintain profitability and build customer trust in a rapidly changing and highly competitive market. In this challenging environment, LOLC Holdings has managed to innovate, manage risks, and establish itself as an industry leader even in the face of economic challenges. This case study presents a compelling study of the interplay between financial performance and stock market behavior with a focus on LOLC Holdings.

Our primary objectives are twofold:

1. Compare the financial performance of LOLC Holdings over its previous years.
2. Analyze how changes in the financial performances of LOLC Holdings impact its stock market performance.

These objectives are crucial for understanding how internal financial decisions and external economic factors influence investor confidence and the market valuation of LOLC Holdings. The study analyzes the company's financial statements, stock market data, and related economic indicators over a ten-year period to provide insights into these dynamics.

2. Literature Review

The relationship between a company's financial performance and its stock market performance has been extensively studied in financial literature. Understanding how financial metrics influence stock prices is critical for both investors and corporate managers in making informed decisions.

Gitman and Zutter (2015) explain that financial statements help assess a company's profitability, liquidity, and stability, providing insights into its long-term growth. Analyzing trends in metrics like revenue, net income, and return on equity (ROE) allows for a meaningful comparison of a company's performance over time. Bodie et al. (2014) highlight that internal strategies and external market conditions also influence profitability, making these factors essential for understanding a company's financial health.

Penman (2013) emphasizes that a company's financial performance drives stock prices, as investors rely on earnings, dividends, and book value to predict stock movements. Positive financial performance tends to boost investor confidence, raising stock prices, while declines can result in lower stock prices. Fama's (1970) Efficient Market Hypothesis (EMH) suggests that stock prices immediately reflect changes in financial performance. For LOLC Holdings, financial fluctuations are expected to directly impact its stock market performance with immediate investor reactions.

3. Methodology

This study aims to compare the financial performance of LOLC Holdings over the past decade and forecast its trajectory for the next five years. Additionally, we examine the impact of financial performance on the fluctuation of LOLC Holdings' stock value in the market. A comprehensive analysis, including ivory valuation, was conducted using LOLC data spanning from 2014 to 2024, with Excel and SPSS employed for analytical purposes.

3.1 Measurements of financial health of LOLC

1. **Trend Analysis** - Trend Analysis is a statistical tool that helps to determine the future movements of a variable on the basis of its historical trends. We evaluated historical and forecasted data from 2014 to 2029. Our trend analysis covered Revenue, Liabilities, Assets and Equity and Net income over time of LOLC Holdings.

2. **Liquidity Analysis** - Liquidity ratios give an idea about a company's ability to convert its assets into cash and pay its current liabilities with that cash whenever required. We evaluated the company's ability to meet its short-term obligations using its current assets.

$$\text{Current Ratio} = \frac{\text{Current Assets}}{\text{Current Liabillities}}$$

Equation 1

3. **Solvency Analysis and Risk Assessment** - Solvency is the ability of a firm to handle longstanding financial obligations. Analysts heavily rely on this metric to assess a company's financial standing, especially its management of long-term debts. Key metrics include the debt-to-equity ratio, which indicates the proportion of debt used relative to equity. This analysis assesses a company's long-term financial stability.

$$\text{Debt to Equity Ratio} = \frac{\text{Total Liabilities}}{\text{Total Equity}}$$

Equation 2

4. **Profitability Analysis** - Profitability ratios can shed light on how well a company's management is operating a business. Investors can use them, along with other research, to determine whether or not a company might be a good investment. This analysis evaluates a company's ability to generate profit relative to its revenue Key metrics include: Gross Profit Margin: Measures the percentage of revenue exceeding the cost of goods sold (COGS), indicating operational efficiency and pricing strategy, Net Profit Margin: Shows the percentage of revenue remaining after all expenses, reflecting overall profitability and financial health.

$$\text{Gross Profit Margin} = \frac{\text{Gross Profit}}{\text{Revenue}}$$

Equation 3.1

$$\text{Net Profit Margin} = \frac{\text{Net Income}}{\text{Gross Income}}$$

Equation 3.2

5. **Efficiency Analysis** - Efficiency Ratios are a measure of how well a company is managing its routine affairs. Conceptually, these ratios analyze how well a company uses its assets to generate revenue. The key metric is the Asset Turnover Ratio (ATR), which measures the efficiency of asset utilization in producing sales.

$$\text{Total Assets Turnover} = \frac{\text{Total Average Sales/ Total Sales}}{\text{Total Average Assets/Total Assets}}$$

Equation 4

3.2 Impact on Stock Market Performance

The financial performance of LOLC Holdings significantly influences its stock market behavior. A historical stock value analysis from 2014 to 2024 was conducted, resulting in a predictive model for stock prices forecasting the trend to 2027 with a 95% confidence interval.

3.2.1 Predictive Model: Multivariate Exponential Growth

Given the complexities of financial modeling, particularly with unpredictable investments like stock returns, we developed a predictive model based on the Multivariate Exponential Growth Model. The model is expressed as:

Stock Price = Exp (4.405709 - 0.824660*Gross profit margin +6.6954E-10*Market capital + 1.2277E-9*Total equity)

Model 1

This model captures the relationship between LOLC's financial metrics and its stock price, providing a robust framework for predicting future trends.

3.2.2 Comparison of Pre-Pandemic and Post-Pandemic Stock Price Models: Analysis of Predicted stock Prices using Pre-Pandemic Model vs. Actual Stock Prices.

In light of the rapid surge in LOLC's share prices following the COVID-19 pandemic, we developed two distinct models for pre-pandemic and post-pandemic using the same Multivariate Exponential Growth approach to analyze the shifts in market dynamics.

<u>Pre-pandemic Model (2014-2019)</u>

Stock Price = Exp (3.984162 +1.7389E-9*Market Capital -0.019591*Inflation -0.467714*Gross profit Margin)

Model 2.1: Pre pandemic Model (2014-2019) 1

<u>Post pandemic Model (2020-2024)</u>

Stock Price = Exp (4.791132 +1.5708E-8*Revenue -0.119470*Debt to equity Ratio +6.1207E-10*Market Capital)

Model 2.2: Post pandemic Model (2020-2024) 1

These models effectively delineate the differing financial drivers of stock performance before and after the pandemic, offering valuable insights into the evolving factors that have shaped LOLC's market valuation.

- Model Analysis: Identify key variables that influenced stock prices during the pre- and post-pandemic periods.
- Residual Plot Analysis: Analyze significant deviations, particularly around the time the pandemic began.
- Box Plot Analysis: Visually compare the distribution and variability of stock prices across the pre- and post-pandemic periods.
- Scatter Plot Analysis: Evaluate the relationship between stock prices predicted by the pre-pandemic model and the actual stock prices observed after the COVID-19 pandemic.

4. <u>Results and Discussions</u>

Trend Analysis

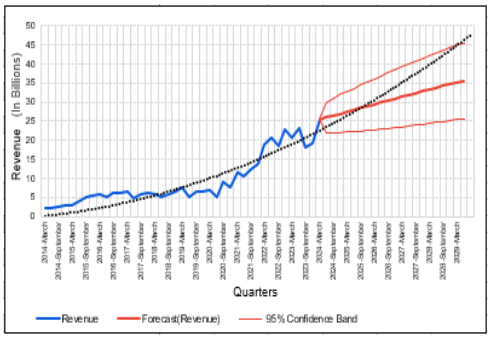

Figure 1.1: Revenue Over Time: March 2014 to Forecasted March 2029.

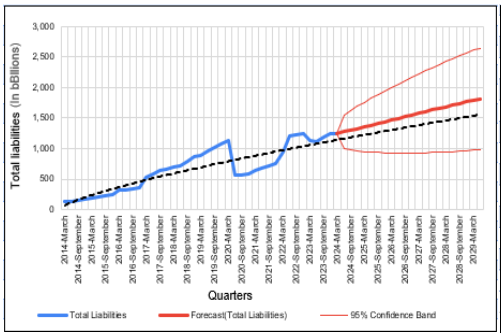

Figure 1.2: Liability Over Time: March 2014 to Forecasted March 2029

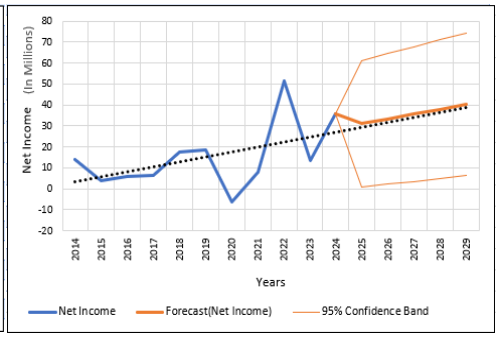

Figure 1.3: Net Income Over Time: March 2014 to Forecasted March 2029

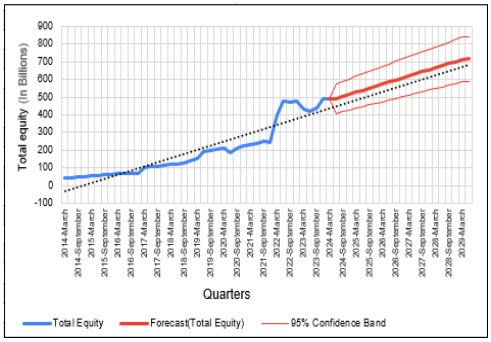

Figure 1.4: Equity Over Time: March 2014 to Forecasted March 2029

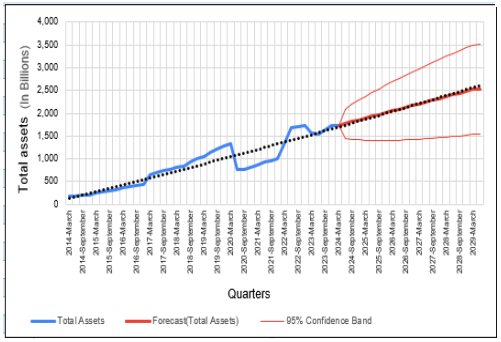

Figure 1.5: Assets Over Time: March 2014 to Forecasted March 2029

Trend Analysis

Discussion - LOLC Holdings has demonstrated consistent growth across revenue, assets, equity, and net income, reflecting a robust financial position. Despite the rising liabilities, which suggest leveraged expansion, the company's growing assets and revenue maintain a healthy balance, indicating managed financial risk. The forecasted data supports this positive outlook, projecting continued upward trends in all key financial indicators through 2029. This positions LOLC Holdings for sustained growth and profitability in the long term, with a strong financial foundation to support future expansion.

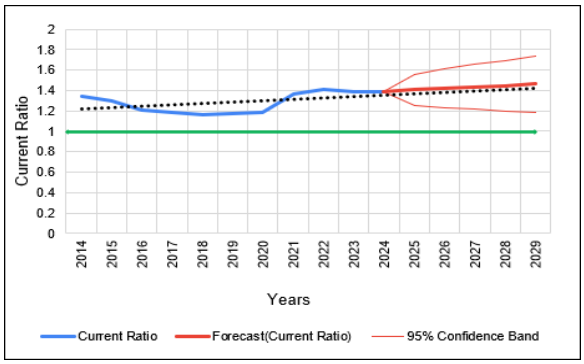

Figure 2: Current Ratio Over Years

Liquidity Analysis

Discussion - LOLC Holdings demonstrates strong liquidity, with its current ratio forecasted to stay above 1.2 over the next five years. The current ratio being always greater than 1.2 means that for every 1 rupee of short-term debt, LOLC has 1.2 or more rupees available to repay their short-term obligations, indicating a strong financial position. This indicates effective liquidity management and a solid financial position, suggesting a positive outlook for the company's ability to meet short-term obligations.

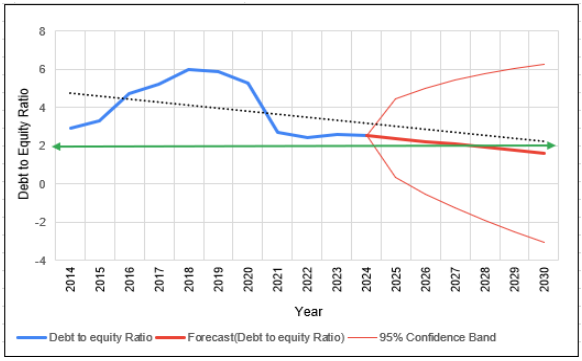

Figure 3: Debt to Equity Ratio Over Years

Solvency Analysis and Risk Assessment

Discussion - The industry leverage benchmark for the finance sector is 2. LOLC Holdings has a large financial sector Called as "LOLC Finance". However, LOLC Holdings shows 2.55 leverage even in 2024, meaning for every 1 rupee of equity, LOLC has 2.55 rupees in long-term debt. But even LOLC Holdings shows a positive trend in solvency with a declining debt-to-equity ratio, expected to stabilize around 2.55 by 2024 and decrease further. This trend indicates improved financial health, reduced financial leverage, and prudent management, positioning the company for future stability and growth.

Profitability Analysis

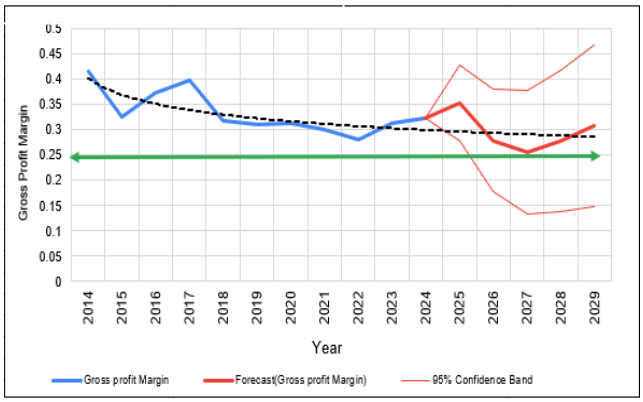

Figure 4.1: Gross Profit Margin

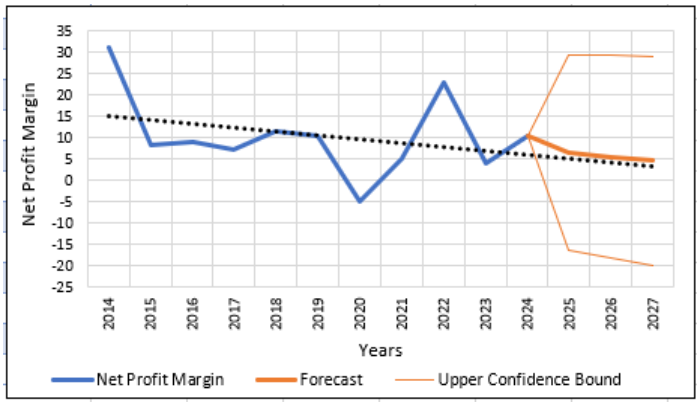

Figure 4.2: Net Profit Margin

Discussion -LOLC Holdings exhibits strong profitability, with a gross profit margin consistently above 25%, signaling efficient operations and solid pricing strategies. That means for every 1 rupee earned by LOLC Holdings, it retains 0.25 rupees after covering the direct costs of production. The net profit margin, despite some fluctuations, remains relatively healthy with a forecasted moderate decline by 2024, indicating a need for improved operational efficiency. Overall, LOLC Holdings demonstrates stable financial health with effective management of profitability.

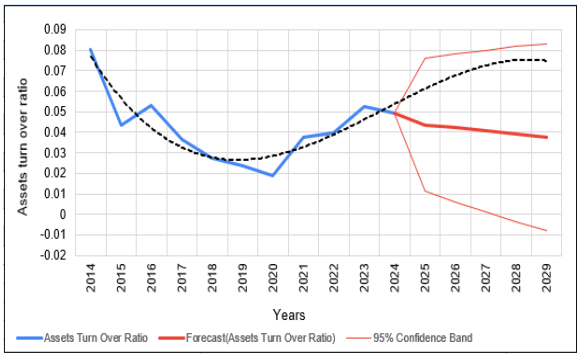

Figure 5: Assets Turn Over Ratio Over Time

Efficiency Analysis

Discussion - LOLC Holdings shows a stable asset turnover ratio in the forecasted period, maintaining a range of 0.04 to 0.05, indicating consistent efficiency in utilizing assets to generate revenue. The slight improvement in 2021 and the upward trend in the forecast suggest potential for enhanced efficiency in the coming years, reflecting positively on the company's operational management.

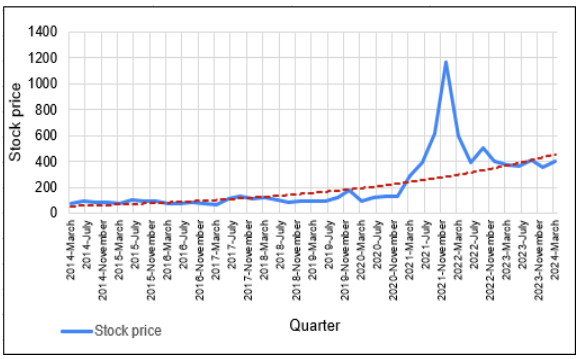

Figure 6: Stock Prices Over Time

Historical Stock Price Analysis

Discussion - LOLC Holdings' stock prices have undergone distinct phases of growth, correction, and stabilization. Starting at LKR 75.00 in March 2014, the stock peaked at LKR 1162.00 in December 2021 before correcting to LKR 392.25 by June 2022. A modest recovery followed, with prices stabilizing between LKR 355.25 and LKR 397.50 in early 2024. This stabilization suggests that the stock has found a new equilibrium, indicating resilience and potential for future growth. The analysis underscores the importance of monitoring market conditions and company performance in pre and post COVID-19 pandemic periods for informed investment decisions.

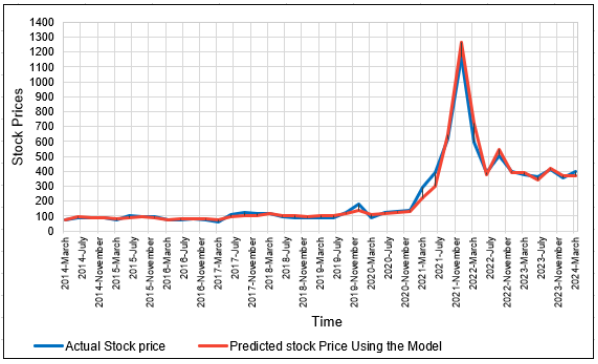

Figure 7: Actual Stock Prices Vs Predicted Stock Prices using Model 01 Over Time

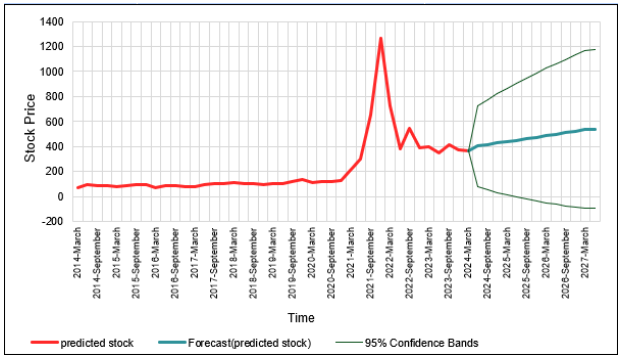

Figure 8: Stock Prices using The Model

Model 1

Model 1 Summary					
					Change Statistics
Model	R	R-Square	Adjusted R Square	St. Error of the Estimate	Sig. F Change
1	0.988	.977	.975	.1233684710	.000
a. Predictors: (Constant), Total Equity, Gross profit Margin, Market Capital b. Dependent Variable: ln(stock)					

Table 1

Model	Coefficients for the Model	Coefficient St. Error	Standardized Coefficient Beta	T Value	Significant Value	95.0% Confidence Lower Bound	Correlation Partial	Statistics VIF (Checking Multicollinearity)
Constant	4.406	.112		39.357	.000	4.179		
Gross profit Margin	-.825	.294	-.075	-2.804	.008	-1.421	-.419	1.123
Market Capital	6.695E-10	.000	.775	20.940	.000	.000	.960	2.158
Total Equity	1.228E-9	.000	.238	6.443	.000	.000	.727	2.148

Table 2

Discussion: The forecast predicts a generally upward trend in LOLC Holdings' stock prices, increasing from around 369.47 in March 2024 to approximately 539.68 by March 2027.

Pre-Pandemic Model

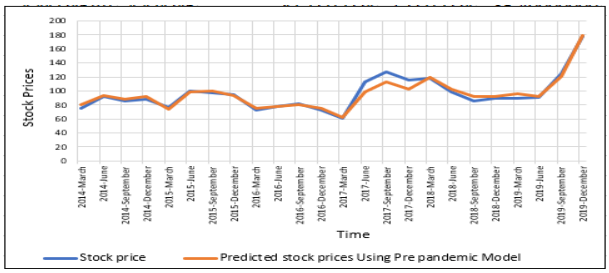

Figure 9: Actual Stock Prices Vs Predicted Stock Prices in Pre-Pandemic Model 2.1 Over Time

Model 2.1: Pre pandemic Model (2014-2019) 1

R-Squared Value – 0.973, Standard Error – 0.0622

Discussion - Pre-COVID-19: Stock prices were more influenced by profitability metrics such as Gross Profit Margin and external economic factors like Inflation. This indicates that during stable economic periods, internal efficiency and broader economic conditions were key determinants of stock price performance.

Post Pandemic Model

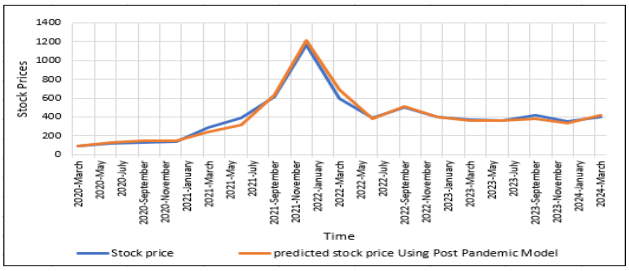

Figure 10: Actual Stock Prices Vs Predicted Stock Prices in Pandemic and Post Pandemic Model 2.1 Over Time

Model 2.2: Post pandemic Model (2020-2024) 1

R – Squared Value – 0.98, Standard Error – 0.105

Discussion - Post-COVID-19: The focus shifted towards Revenue and the Debt-to-Equity Ratio, suggesting that investors became more concerned with growth potential and financial stability after COVID-19. This confirmed by a significant revenue increment in the plantation sector of LOLC Holdings in 2021, as shown in the financial statements. Investors shifted their focus to how much money the company was making and its ability to manage debt. Concerns about how companies would survive in a challenging economic environment. This change reflects the market's adaptation to a new economic environment where revenue generation and financial health became critical indicators of a company's value.

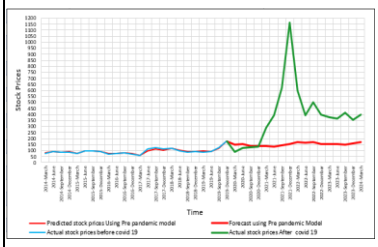

Figure 11: Difference between Actual stock price for post COVID-19 and forecasted stock price using pre pandemic model

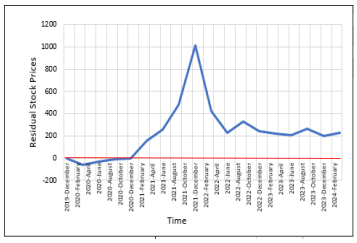

Figure 12: Residuals In-between pre pandemic model forecasted stock prices and Actual stock prices

Residual Plot Analysis

Discussion - The residual plot reveals significant deviations around the onset of the COVID-19 pandemic, highlighting its profound impact on stock prices. The sharp rise and subsequent dip in residuals during late 2020 to early 2021 indicate the pandemic's role as a "black swan" event, causing unpredictable market shifts that no pre-pandemic model could have anticipated. This underscores the pandemic's lasting effects on market dynamics and the importance of adapting models to account for such unprecedented events.

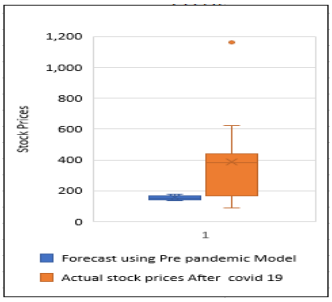

Figure 13: Box Plot for Predicted Stock Prices and Actual Stock Prices during and after pandemic.

Forecast using Pre pandemic Model		Actual stock prices After COVID-19	
Mean	154.4645548	Mean	384.4888889
Standard error	3.144037253	Standard error	58.72974029
Median	153.0263159	Median	383.625
Standard Deviation	13.33902037	Standard Deviation	249.1691857
Sample Variance	177.9294645	Sample Variance	62085.2831
Kurtosis	-0.868908239	Kurtosis	4.931476416
Skewness	0.18311045	Skewness	1.751303712
Range	44.35116005	Range	1071.2
Count	18	Count	18

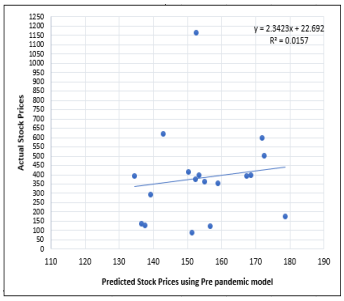

Figure 14: Scatter Plot for Predicted Stock Prices using model and actual Stock Prices during and after pandemic.

Box Plot and Scatter Plot Analysis

Discussion - The analysis reveals that the pre-pandemic model, while accurate for its time, fails to predict the substantial increase and volatility in LOLC Holdings' stock prices post-pandemic. The box plot shows a significant rise in stock price variability, and the scatter plot indicates a weak correlation between predicted and actual prices, with an R-squared value of only 0.0157. This underscores that the pandemic introduced new market dynamics, rendering the pre-pandemic model inadequate for the current economic environment, highlighting the need for updated modeling approaches.

5. Conclusion

The study aims to measure the financial health of LOLC Holdings, assess its stock health, and analyze LOLC Holdings' performance in the stock market.

The trend analysis from 2019 to 2024, along with forecasts through 2029, shows a positive financial trajectory for LOLC Holdings, with consistent growth in revenue, liabilities, equity, total assets, and net income. The company's effective management of rising liabilities, coupled with increased revenue and net income, highlights robust financial health and operational efficiency, demonstrating its ability to generate sufficient earnings and sustain growth. From 2014 to 2029, LOLC Holdings has consistently maintained a current ratio above 1, indicating strong liquidity. The declining debt-to-equity ratio highlights improved financial stability through increased equity and reduced leverage. Despite fluctuations, the company's healthy gross profit margin and forecasts of enhanced profitability signal robust financial health. Although the asset turnover ratio has declined, a slight recovery is anticipated, suggesting potential efficiency improvements. So, we can conclude that, LOLC Holdings presents a strong investment opportunity with a solid financial health and potential for sustained profitability until 2029.

The five-year stock price forecast for LOLC Holdings shows an upward trend, reflecting positive market sentiment, expected financial growth, and strategic success. However, historical stock prices have experienced significant fluctuations due to the COVID-19 pandemic, which has substantially impacted LOLC's stock market performance. Pre-pandemic, stock prices were primarily driven by profitability factors such as market capitalization, inflation, and gross profit margin. Post-pandemic, however, stock prices are increasingly influenced by financial stability indicators like revenue and the debt-to-equity ratio and also market capital. This shift has made traditional pre-COVID forecasting models less effective, as they fail to account for the substantial economic and market changes brought about by the pandemic. Consequently, stock price volatility has increased in the post-pandemic period. Despite these fluctuations, the stock has recently stabilized, indicating a potential new equilibrium and the possibility for future growth. Overall, LOLC Holdings' stock health remains robust, but continuous monitoring and updated strategies are essential to navigate the evolving market conditions.

LOLC Holdings presents a strong investment opportunity, with resilient financial health and effective debt management. Consistent growth and a positive profitability outlook position the company well for solid returns.

6. References

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Contribution

s16201 Dissanayaka K. D. J. S.

My contribution to the final report focused on crafting the conclusion, and summarizing key findings on LOLC Holdings' financial performance and stock market health. Additionally, I was responsible for compiling and formatting the references in APA style, ensuring accurate citation of all sources used throughout the report, thereby supporting the clarity and consistency of the document.

s16326 Liyanage S. S.

As my Contribution for the final report, I created the cover page, as well as the results and discussion section. This included preparing and designing all the figures and writing the accompanying results and discussion. Additionally, I formatted the entire document and structured the final report layout. Essentially, I handled both the visual and content organization of the report to ensure it was well-presented.

s16387 Kodikara P. G.

For this report, I handled the Introduction and Literature Review. The Introduction outlines LOLC Holdings and the objectives of the case study. The Literature Review covers research on how financial performance affects stock prices, including Fama's hypothesis, and factors influencing financial health. These sections provide the foundation for analyzing LOLC's data.

s16389 Jayasinghe H. M. S. C.

My responsibility in this report was to thoroughly document the methodology section, ensuring it was clear and detailed. I included all relevant equations and models used throughout the study, explaining their significance. Additionally, I linked each equation and model to its respective section for easy reference. This helped maintain clarity and consistency in the report's overall structure.